Executive Summary: Secure Cloud Storage Environment

This report details the design, implementation, and security evaluation of a containerized cloud storage architecture. The environment utilizes MinIO, deployed via Docker, to create a secure, S3-compatible object storage platform. A simulated dataset of 100,000 corporate records was generated to represent Human Resources and Finance department assets.

The project enforced strict security controls across multiple layers. OpenSSL was used to generate self-signed TLS/SSL certificates, guaranteeing encryption of data in transit. Data at rest was secured using Server-Side Encryption (SSE-S3) integrated with a Key Management Service (KMS). Access to sensitive data was restricted using Role-Based Access Control (RBAC), implementing the principle of least privilege through isolated departmental buckets and user-specific JSON policies.

We executed a rigorous six-part security testing matrix. These tests validated the enforcement of RBAC boundaries, the rejection of unauthenticated requests, the mandatory use of HTTPS, the active state of SSE-S3 encryption, and the successful capture of administrative audit logs for unauthorized access attempts.

While the current architecture successfully demonstrates fundamental cybersecurity principles, isolated container deployments present single points of failure. Future enterprise scaling recommendations include migrating to a Kubernetes-orchestrated cluster for high availability, integrating Active Directory for centralized identity management, and utilizing Elasticsearch for persistent security information and event management (SIEM).